

Probability & Statistics

Course Code: MATH-6608

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PROJECT REPORT



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1. Topic & Introduction

Topic: Healthy eating habits among young adults

Introduction

Healthy eating habits among young adults play a crucial role in shaping long-term physical and mental well-being. This stage of life is often characterized by increased independence, academic pressure, and lifestyle changes, which can significantly influence dietary patterns. Many young adults tend to adopt irregular eating habits due to busy schedules, social influences, and easy access to fast food.

Understanding these habits is essential to identify nutritional gaps and promote healthier lifestyles. This study aims to analyze the eating behaviors, preferences, and influencing factors among young adults, highlighting how demographic characteristics, economic conditions, and lifestyle choices affect their dietary decisions.

2. Methodology & MCQs

The project utilized a structured questionnaire of 19 core Multiple Choice Questions (MCQs). Data was collected from 87 student respondents and analyzed using Python libraries for statistical computation.

1. How often do you eat fruits?

- Daily
- Sometimes
- Rarely

2. How often do you eat vegetables?

- Daily
- Occasionally
- Never

3. How many meals do you eat per day?

- 1–2 meals
- 3 meals

- More than 3 meals

4. Do you skip breakfast?

- Always
- Sometimes
- Never

5. How often do you eat fast food?

- Frequently
- Occasionally
- Rarely

6. How much water do you drink daily?

- Less than 1 liter
- 1–2 liters
- More than 2 liters

7. Do you read nutrition labels on food?

- Always
- Sometimes
- Never

8. What influences your food choices the most?

- Taste
- Price
- Health

9. How often do you consume sugary drinks?

- Daily
- Occasionally
- Rarely

10. Do you follow a balanced diet?

- Yes
- Sometimes
- No

11. How often do you eat home-cooked meals?

- Daily
- Sometimes
- Rarely

12. Do you think your diet is healthy?

- Yes
- Maybe
- No

13. How often do you snack between meals?

- Frequently
- Occasionally
- Rarely

14. Do you consume junk food when stressed?

- Yes
- Sometimes
- No

15. How aware are you about healthy eating?

- Very aware
- Somewhat aware
- Not aware

16. How often do you eat late at night?

- Often
- Sometimes
- Never

17. Do you include protein in your daily diet?

- Yes
- Sometimes
- No

18. What type of food do you prefer?

- Healthy food
- Fast food
- Both equally

19. Do you exercise regularly?

- Yes
- Sometimes
- No

3. Code and output of analysis

I. Mean and Median of Variables

```
# --- SECTION 1: MEASURES OF LOCATION (week 2) ---
# TARGET QUESTION: "Survey Rating"

# 1. Calculate the values
mean_val = df['rating'].mean()
median_val = df['rating'].median()
mode_val = df['rating'].mode()[0]

print(f" Results for 'Survey Rating'")
print(f"Mean: {mean_val:.2f} | Median: {median_val}")

# 2. Visualizing with clean markers
plt.figure(figsize=(10, 6))
sns.countplot(x='rating', data=df, palette='viridis')

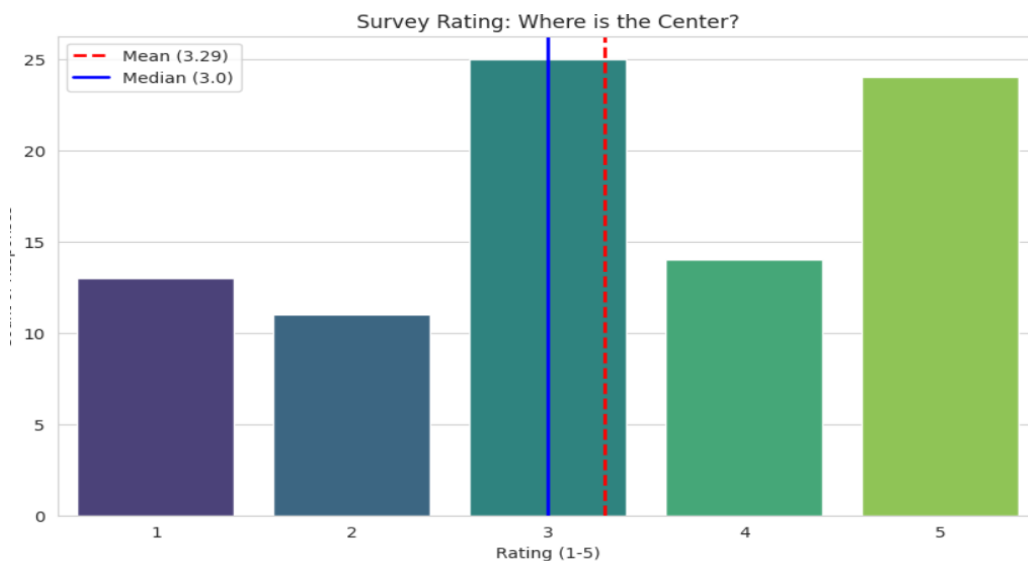
# --- PROOF LINES ---
# We add vertical lines to SHOW where the Mean and Median are.
# Note: Since the chart starts at index 0, we subtract 1 from our values to align them.

# Red dashed line for Mean
plt.axvline(x=mean_val-1, color='red', linestyle='--', linewidth=2, label=f'Mean ({mean_val:.2f})')

# Blue solid line for Median
plt.axvline(x=median_val-1, color='blue', linestyle='-', linewidth=2, label=f'Median ({median_val})')

plt.title('Survey Rating: Where is the Center?')
plt.xlabel('Rating (1-5)')
plt.ylabel('Count of Responses')
plt.legend() # This shows the "key" so you know which line is which
plt.show()
```

Output box



II. Measure of Dispersion

```
# Creating a numeric version of the column
df['meals_numeric'] = df['meals_per_day'].map(mapping)

# 2. Calculating the "Spread" (Dispersion)
# Range = The gap between the smallest and largest value
meal_range = df['meals_numeric'].max() - df['meals_numeric'].min()

# Variance = How much the data points differ from the average
meal_variance = df['meals_numeric'].var()

# Standard Deviation = The average distance from the mean (most common way to show spread)
meal_std = df['meals_numeric'].std()

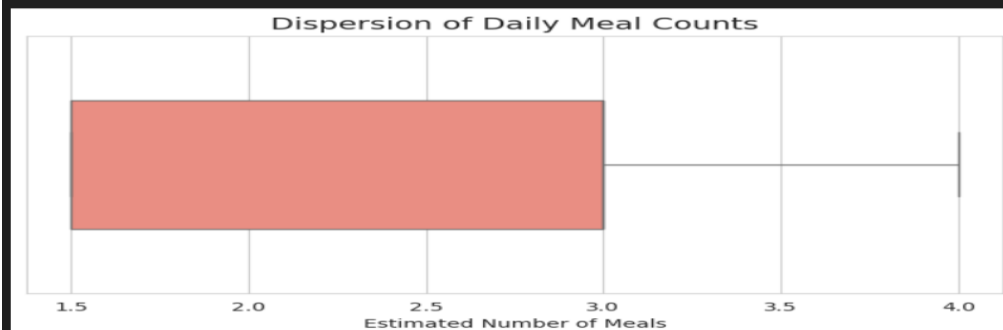
# Printing results clearly
print(f"--- Stats for: 'How many meals do you eat per day?' ---")
print(f"-----")
print(f"Range: {meal_range} meals")
print(f"Variance: {meal_variance:.2f}")
print(f"Standard Deviation: {meal_std:.2f}")

# 3. Visualization: Box Plot
# The 'box' shows where most people fall, and the 'whiskers' show the outliers.
plt.figure(figsize=(8, 4))
sns.boxplot(x=df['meals_numeric'], color='salmon', width=0.5)

plt.title('Dispersion of Daily Meal Counts', fontsize=14)
plt.xlabel('Estimated Number of Meals')
```

Output box

```
Range: 2.5 meals
Variance: 1.00
Standard Deviation: 1.00
```



III. Basic conditional probability

```
total_people = len(df)
always_skip_count = len(df[df['skip_breakfast'] == 'Always'])
prob_always_skip = always_skip_count / total_people

print(f"--- Basic Probability ---")
print(f"P(Always Skip Breakfast): {prob_always_skip:.2%}")

# 2. Conditional Probability: If someone ALWAYS skips breakfast,
# what is the probability they ALSO say "Yes" to stress eating junk food?
# Formula: P(Junk="Yes" | Skip="Always")

# First, find people who do BOTH
both_count = len(df[(df['skip_breakfast'] == 'Always') & (df['stress_eating'] == 'Yes')])

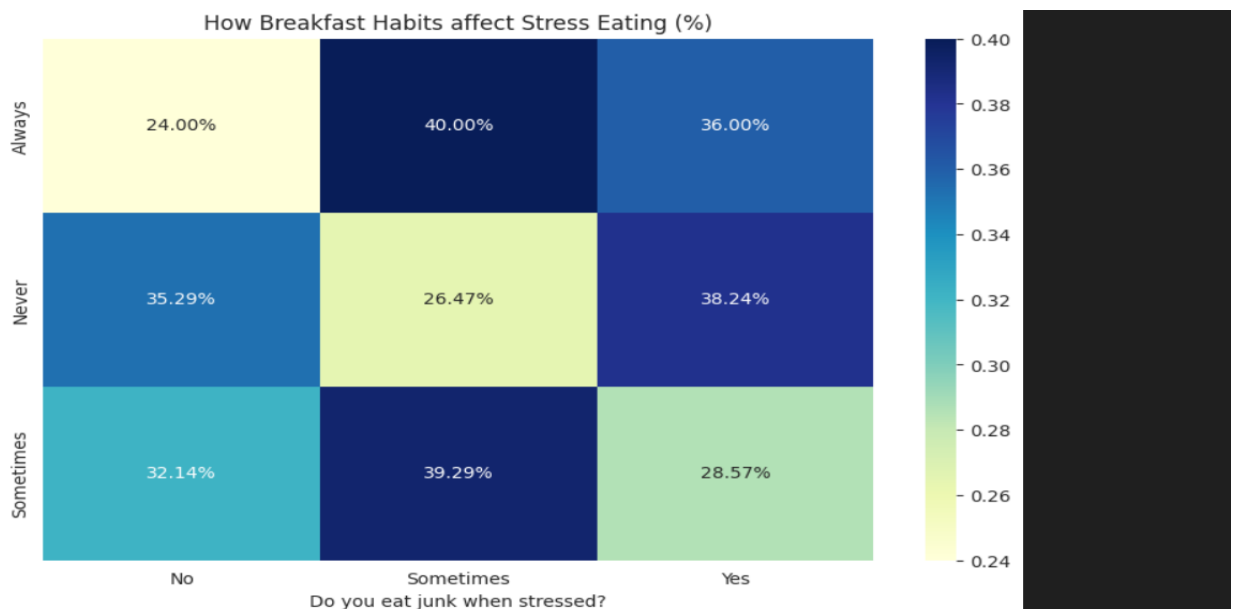
# Calculate the probability based only on the "Always Skippers" group
cond_prob = both_count / always_skip_count if always_skip_count > 0 else 0

print(f"\n--- Conditional Probability ---")
print(f"P(Eat Junk when Stressed | Always Skip Breakfast): {cond_prob:.2%}")

# --- VISUALIZATION ---
# This "Heatmap" shows the probabilities for every combination!
# Higher percentages mean a stronger "connection" between habits.
ct = pd.crosstab(df['skip_breakfast'], df['stress_eating'], normalize='index')
plt.figure(figsize=(10, 6))
sns.heatmap(ct, annot=True, cmap='YlGnBu', fmt='.2%')

plt.title('How Breakfast Habits affect Stress Eating (%)')
plt.ylabel('Do you skip breakfast?')
plt.xlabel('Do you eat junk when stressed?')
plt.show()
```

Output box



IV. Ramdom Variable

```
prob_dist = df[ 'meals_numeric' ].value_counts(normalize=True).sort_index()

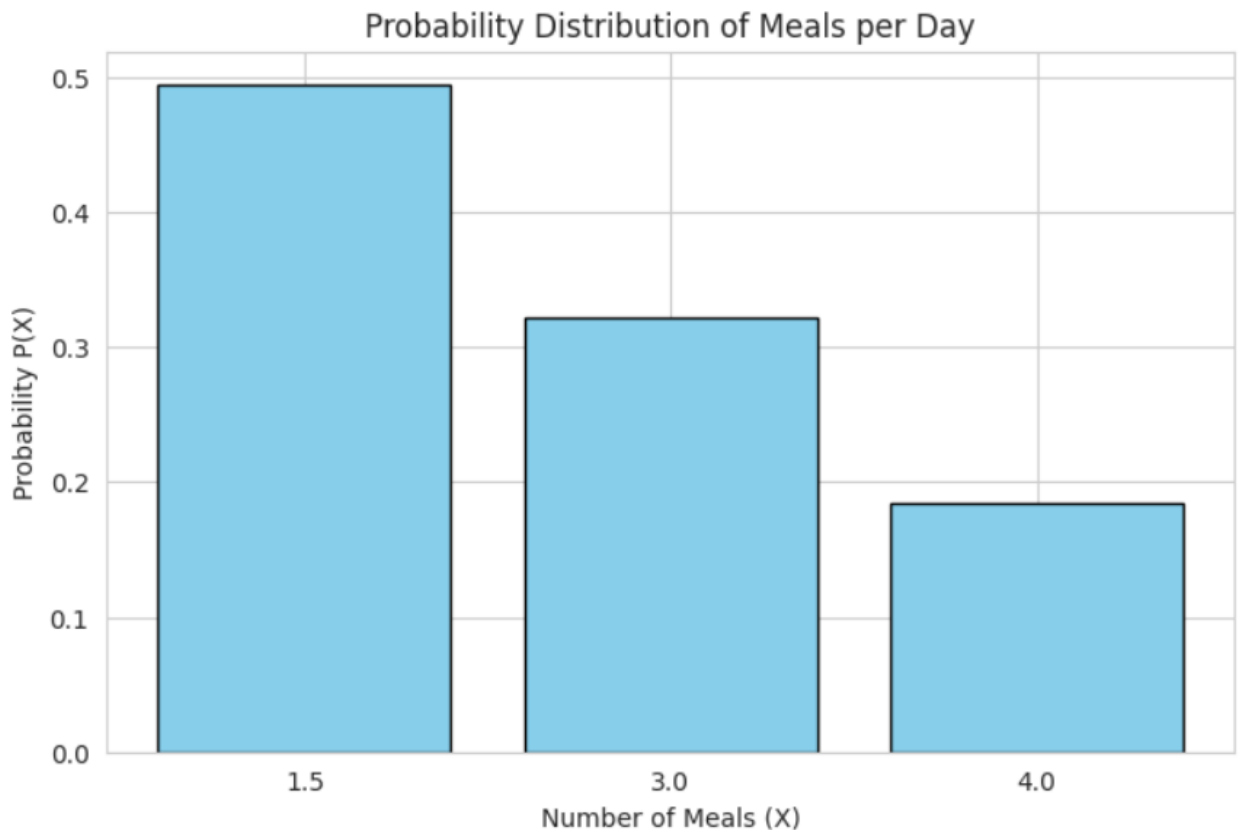
# 2. Calculate the Mean (Expected Value)
# Formula:  $E[X] = \sum (x * P(X=x))$ 
mean_x = sum(prob_dist.index * prob_dist.values)

# 3. Calculate the Variance
# Formula:  $Var(X) = E[X^2] - (E[X])^2$ 
mean_x_sq = sum((prob_dist.index**2) * prob_dist.values)
variance_x = mean_x_sq - (mean_x**2)

print("--- Probability Distribution of X (Meals) ---")
print(prob_dist)
print("\n--- Results ---")
print(f"Expected Value (Mean  $E[X]$ ): {mean_x:.2f}")
print(f"Variance ( $Var(X)$ ): {variance_x:.2f}")

# VISUALIZATION
# This shows the Probability Mass Function (PMF)
plt.figure(figsize=(8, 5))
plt.bar(prob_dist.index.astype(str), prob_dist.values, color='skyblue', edgecolor='black')
plt.title('Probability Distribution of Meals per Day')
plt.xlabel('Number of Meals (X)')
plt.ylabel('Probability P(X)')
plt.show()
```

Output box



V. Binomial distribution

```
from scipy.stats import binom

# 1. Calculate 'p' (The probability of success from our data)
total_responses = len(df)
yes_count = len(df[df['balanced_diet'] == 'Yes'])
p = yes_count / total_responses

# 2. Set the parameters for our Binomial experiment
n = 10 # We imagine picking a group of 10 random people
k_values = list(range(n + 1)) # Possible number of successes: 0, 1, 2... up to 10

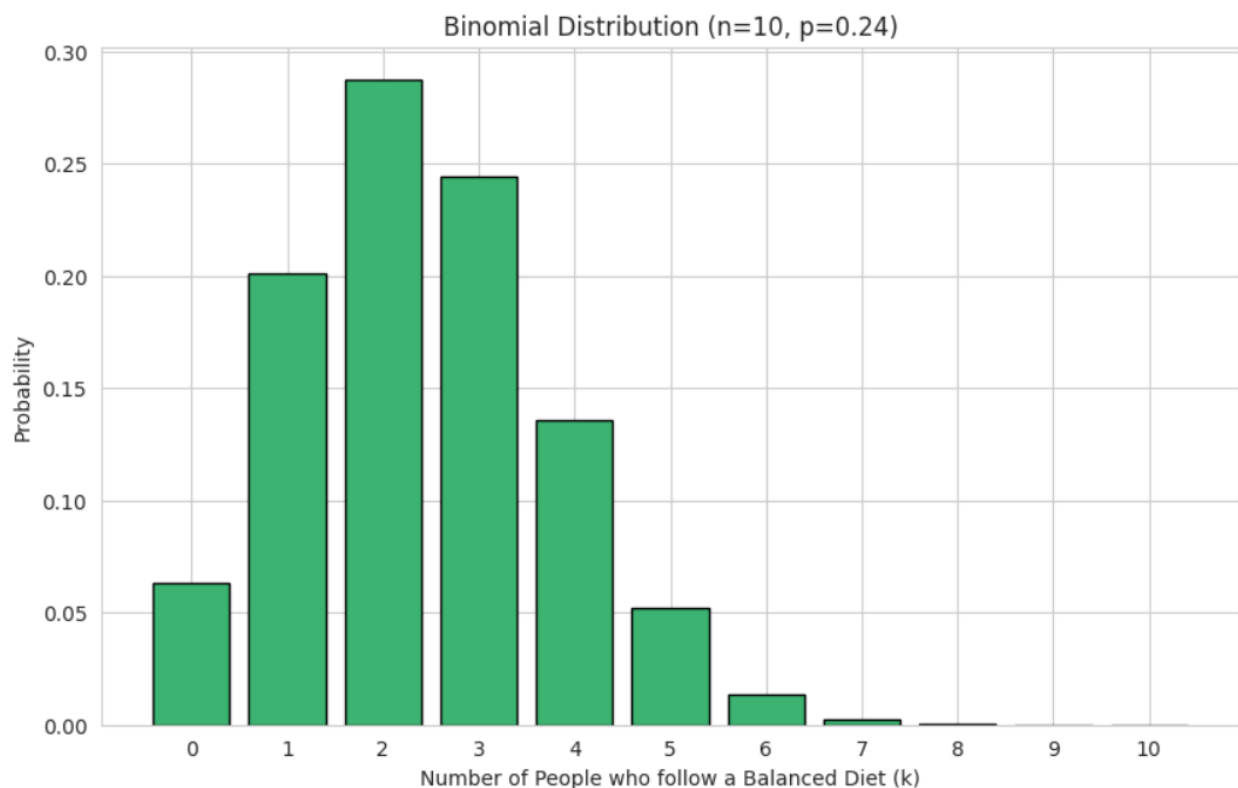
# 3. Calculate the Binomial Probability for each 'k'
# Formula:  $P(X=k) = \binom{n}{k} * (p^k) * ((1-p)^{(n-k)})$ 
probabilities = [binom.pmf(k, n, p) for k in k_values]

print("--- Binomial Distribution Parameters ---")
print(f"Probability of Success (p): {p:.2f}")
print(f"Number of Trials (n): {n}")
print("\n--- Probability of 'k' successes in 10 people ---")
for k, prob in zip(k_values, probabilities):
    print(f"k={k}: {prob:.4f}")

# VISUALIZATION
plt.figure(figsize=(10, 6))
plt.bar(k_values, probabilities, color='mediumseagreen', edgecolor='black')
plt.title(f'Binomial Distribution (n=10, p={p:.2f})')
plt.xlabel('Number of People who follow a Balanced Diet (k)')
plt.ylabel('Probability')
plt.xticks(k_values)
plt.show()
```

Output

box



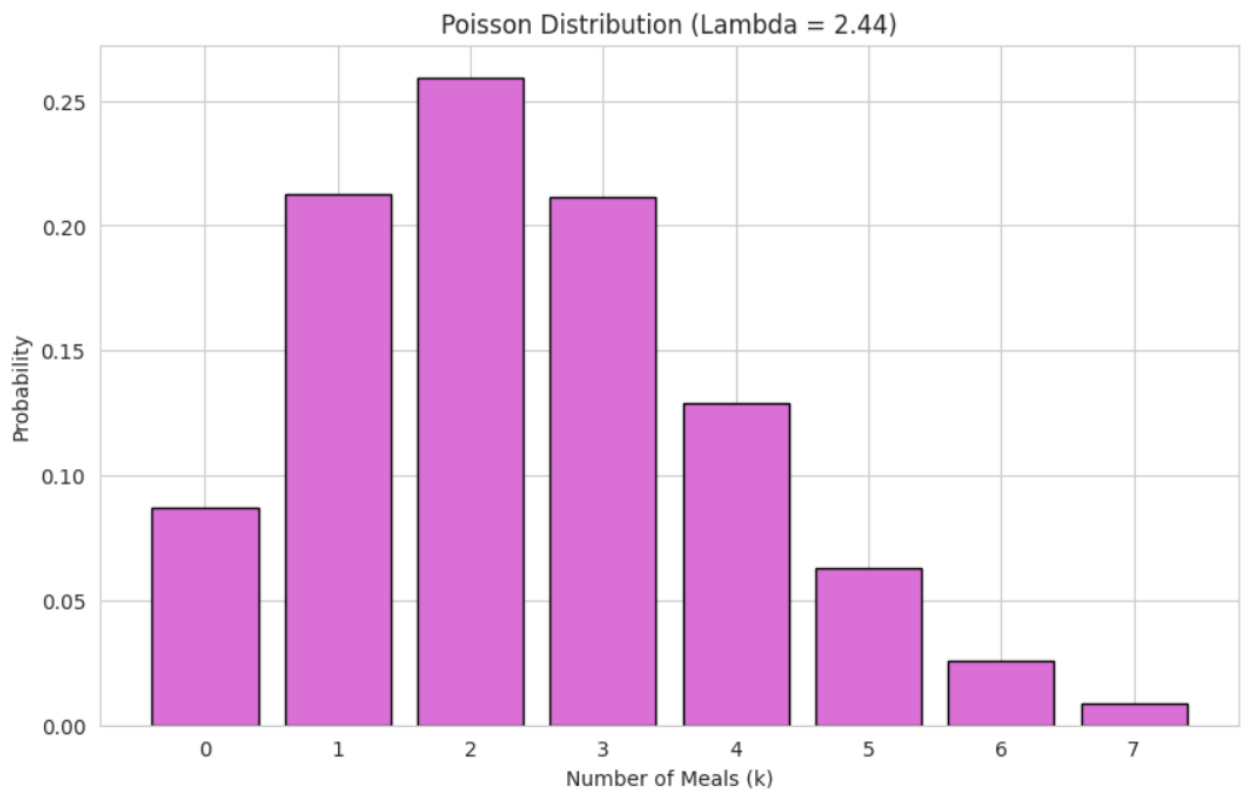
VI. poisson Disturbution

```
poisson_probs = poisson.pmf(k_values, lam)

print("--- Poisson Distribution Results ---")
print(f"Average Rate (Lambda): {lam:.2f} meals per day")
print("\n--- Probability of eating exactly 'k' meals ---")
for k, prob in zip(k_values, poisson_probs):
    print(f"k={k}: {prob:.4f}")

# VISUALIZATION
plt.figure(figsize=(10, 6))
plt.bar(k_values, poisson_probs, color='orchid', edgecolor='black')
plt.title(f'Poisson Distribution (Lambda = {lam:.2f})')
plt.xlabel('Number of Meals (k)')
plt.ylabel('Probability')
plt.xticks(k_values)
plt.show()
```

Output box



VII. Normal Disturbution

```
mu = df['rating'].mean()
sigma = df['rating'].std()

# 2. Create a range of numbers for the x-axis (from min to max rating)
x_axis = np.linspace(df['rating'].min() - 1, df['rating'].max() + 1, 100)

# 3. Calculate the Normal PDF (Probability Density Function)
# This creates the smooth "bell" line based on our mu and sigma
bell_curve = norm.pdf(x_axis, mu, sigma)

print("--- Normal Distribution Summary ---")
print(f"Mean (mu): {mu:.2f}")
print(f"Standard Deviation (sigma): {sigma:.2f}")

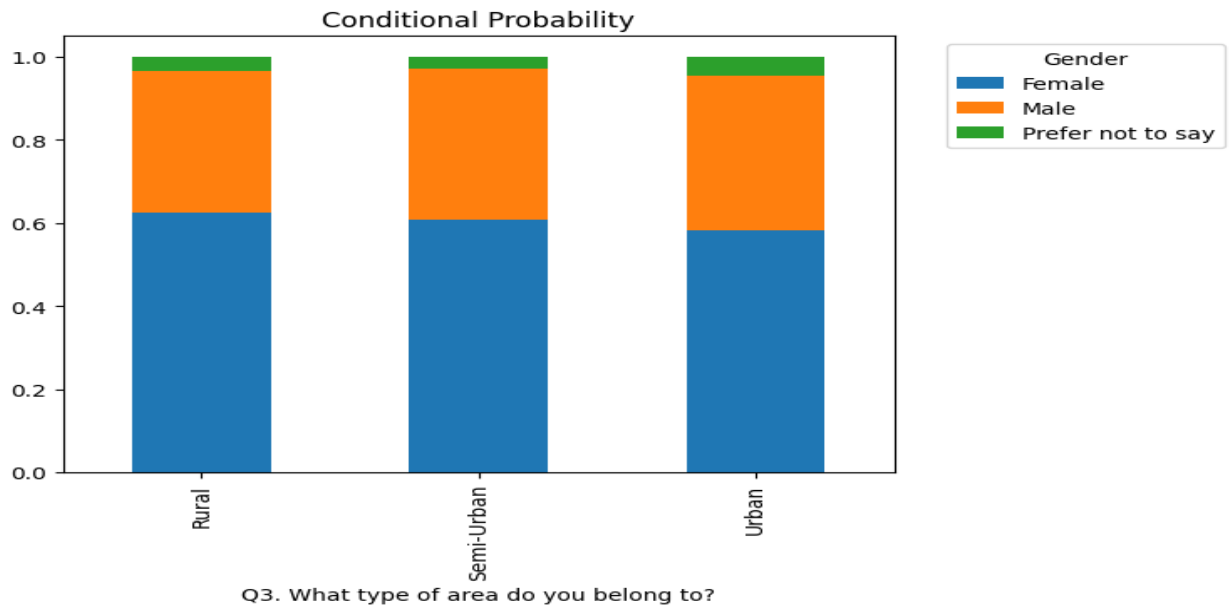
# 4. Plotting the actual data vs the theoretical Normal curve
plt.figure(figsize=(10, 6))

# The histogram shows the actual survey counts
sns.histplot(df['rating'], kde=False, stat="density", color='lightgray', label='Actual Data')

# The line shows the "Normal Distribution" model
plt.plot(x_axis, bell_curve, color='blue', linewidth=3, label='Normal Curve')

plt.title(f'Normal Distribution Model (Mean={mu:.2f}, Std Dev={sigma:.2f})')
plt.xlabel('Survey Rating')
plt.ylabel('Density')
plt.legend()
plt.show()
```

Output box



VIII. Central limit theorem

```
water_mapping = {
    'Less than 1 liter': 0.5,
    '1-2 liters': 1.5,
    'More than 2 liters': 2.5
}
df['water_numeric'] = df['water_intake'].map(water_mapping)

# 2. The Central Limit Theorem Experiment
# we are going to take 1000 random samples of 30 people each
# and calculate the average (mean) for every single sample.
sample_means = []

for i in range(1000):
    # Take a random sample of 30 responses
    sample = df['water_numeric'].sample(n=30, replace=True)
    # Calculate the mean of that sample and save it
    sample_means.append(sample.mean())

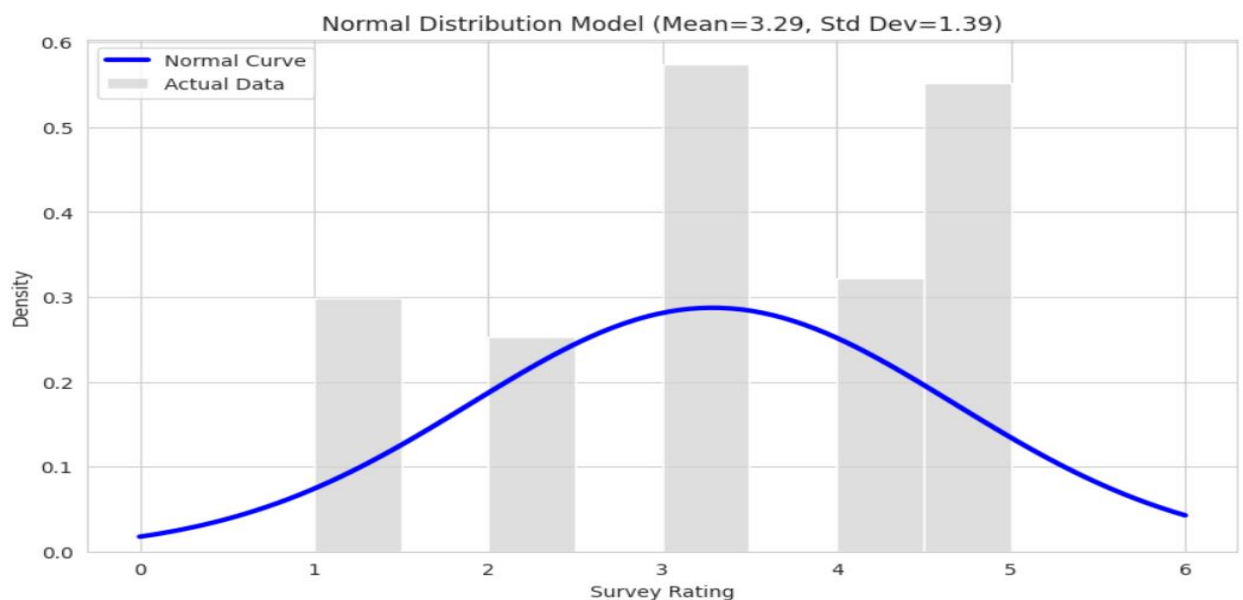
print("--- CLT Summary ---")
print(f"Population Mean: {df['water_numeric'].mean():.2f}")
print(f"Average of Sample Means: {np.mean(sample_means):.2f}")

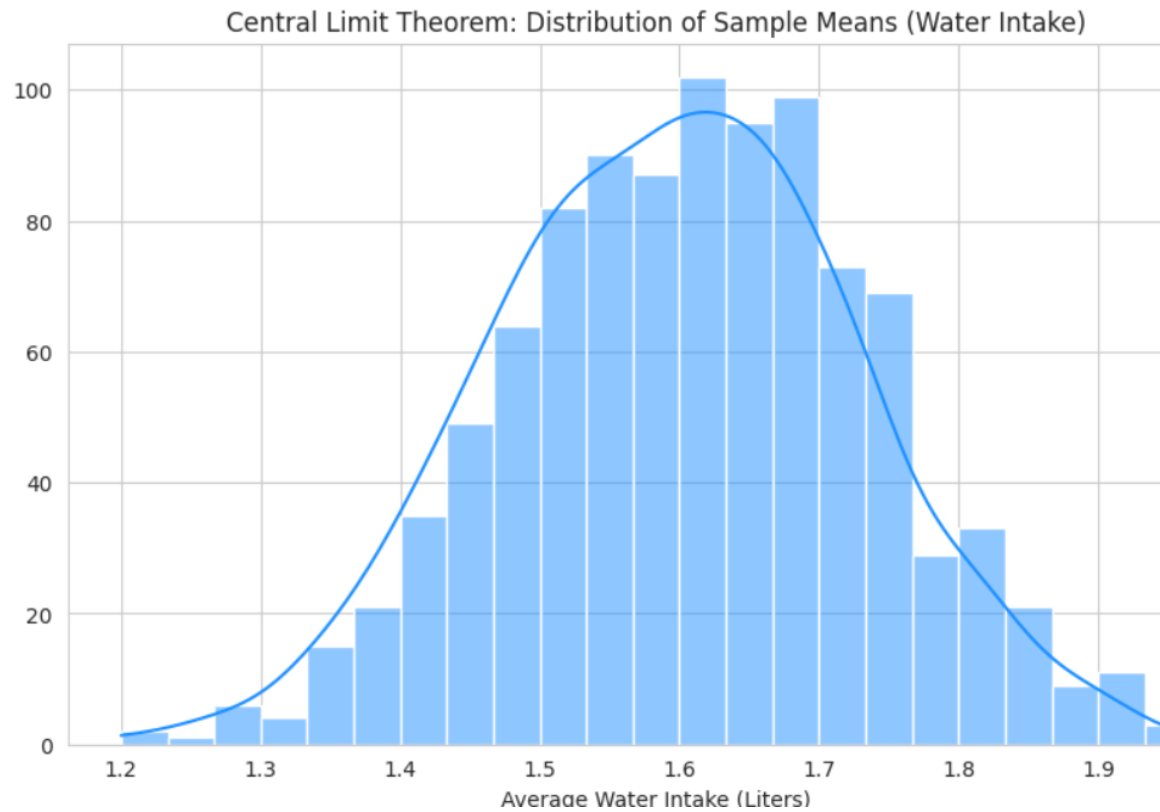
# 3. Visualization
plt.figure(figsize=(10, 6))

# Plotting the distribution of the sample means
sns.histplot(sample_means, kde=True, color='dodgerblue')

plt.title('Central Limit Theorem: Distribution of Sample Means (Water Intake)')
plt.xlabel('Average Water Intake (Liters)')
plt.ylabel('Frequency')
plt.show()
```

❖ Output





IX. Confidence interval

```
awareness_map = {
    'Very aware': 3,
    'Somewhat aware': 2,
    'Not aware': 1
}
df['awareness_numeric'] = df['health_awareness'].map(awareness_map)

# 2. Calculate the stats needed for a 95% Confidence Interval
import scipy.stats as st

data = df['awareness_numeric']
n = len(data)
mean = data.mean()
std_err = st.sem(data) # Standard Error

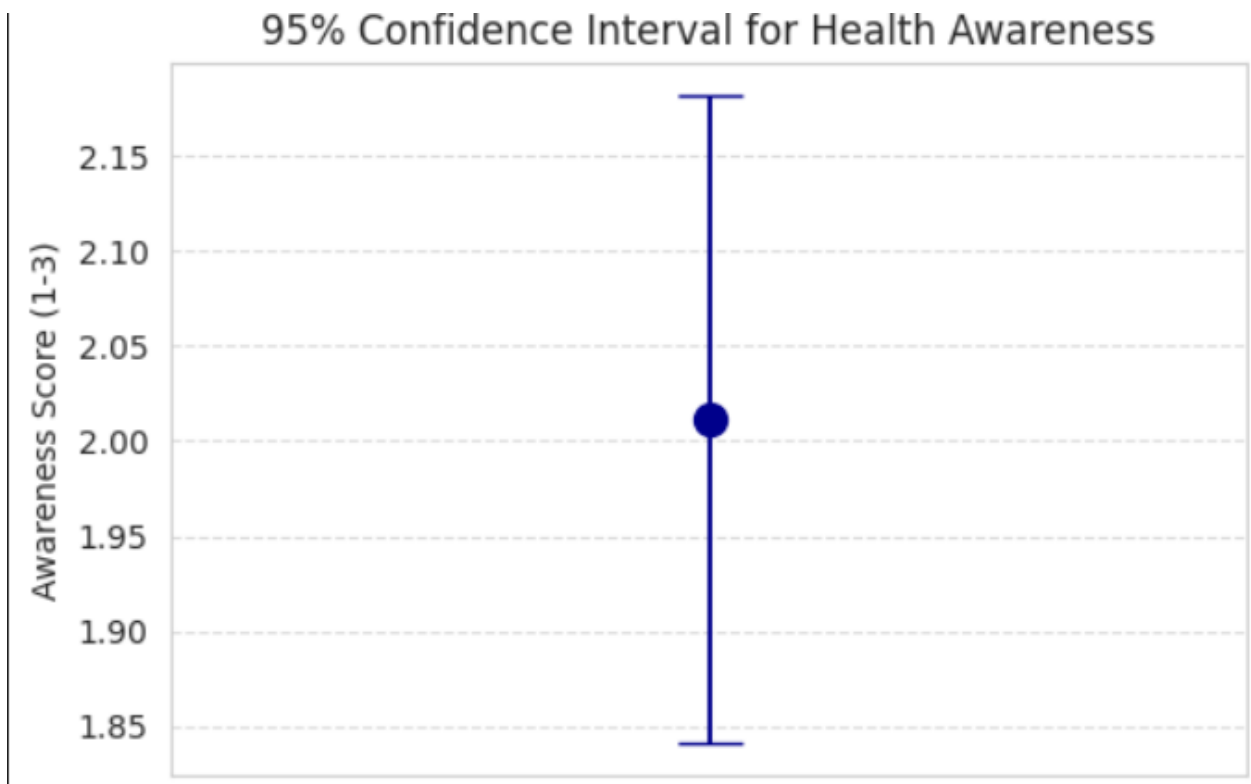
# 3. Calculate the Interval (95%)
# This gives us the 'Lower' and 'Upper' bound
confidence_level = 0.95
interval = st.t.interval(confidence_level, df=n-1, loc=mean, scale=std_err)

# 4. Create a clean table for the output
ci_table = pd.DataFrame({
    'Metric': ['Sample Size (n)', 'Average Awareness Score', '95% Confidence (Lower)', '95% Confidence (Upper)'],
    'Value': [n, round(mean, 2), round(interval[0], 2), round(interval[1], 2)]
})

print("CONFIDENCE INTERVAL RESULTS")
display(ci_table)
```

❖ Output

box



X. Hypothesis Testing

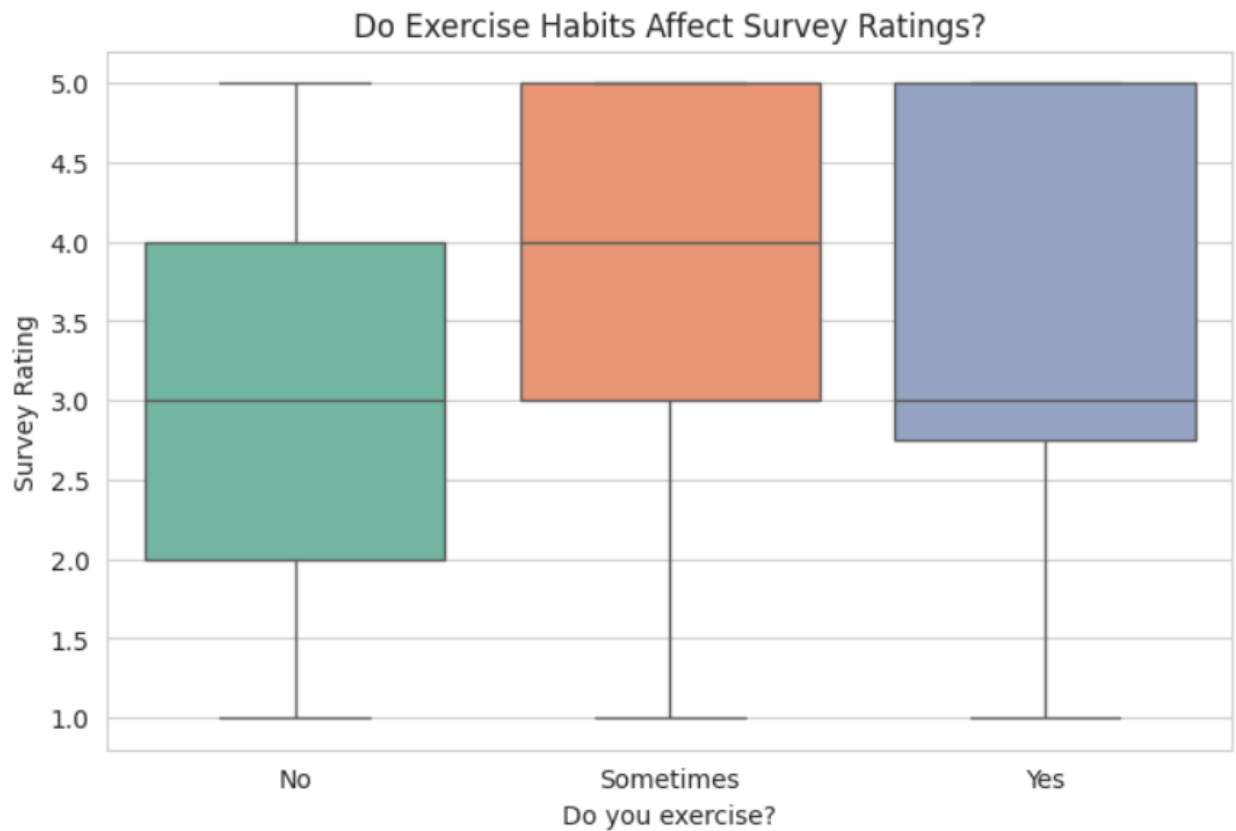
```
hypothesis_table = pd.DataFrame({
    'Metric': ['T-Statistic', 'P-Value', 'Mean Rating (Exercises)', 'Mean Rating (No Exercise)'],
    'Value': [round(t_stat, 4), round(p_value, 4), round(exercises.mean(), 2), round(no_exercise.mean(), 2)]
})

print("TESTING RESULTS")
display(hypothesis_table)

# 4. Decision logic
alpha = 0.05
if p_value < alpha:
    print("\nResult: Significant difference found (Reject H0)")
else:
    print("\nResult: No significant difference found (Fail to reject H0)")

# VISUALIZATION
# Boxplot comparing the two groups
plt.figure(figsize=(8, 5))
sns.boxplot(x='exercise', y='rating', data=df, palette='Set2')
plt.title('Do Exercise Habits Affect Survey Ratings?')
plt.xlabel('Do you exercise?')
plt.ylabel('Survey Rating')
plt.show()
```

❖ Output box



XI. Linear Regression

```
regression_table = pd.DataFrame({
    'Component': ['Slope (m)', 'Intercept (c)', 'Regression Equation'],
    'Value': [round(slope, 4), round(intercept, 4), f"Y = ({round(slope,2)} * X) + {round(intercept,2)}"]
})

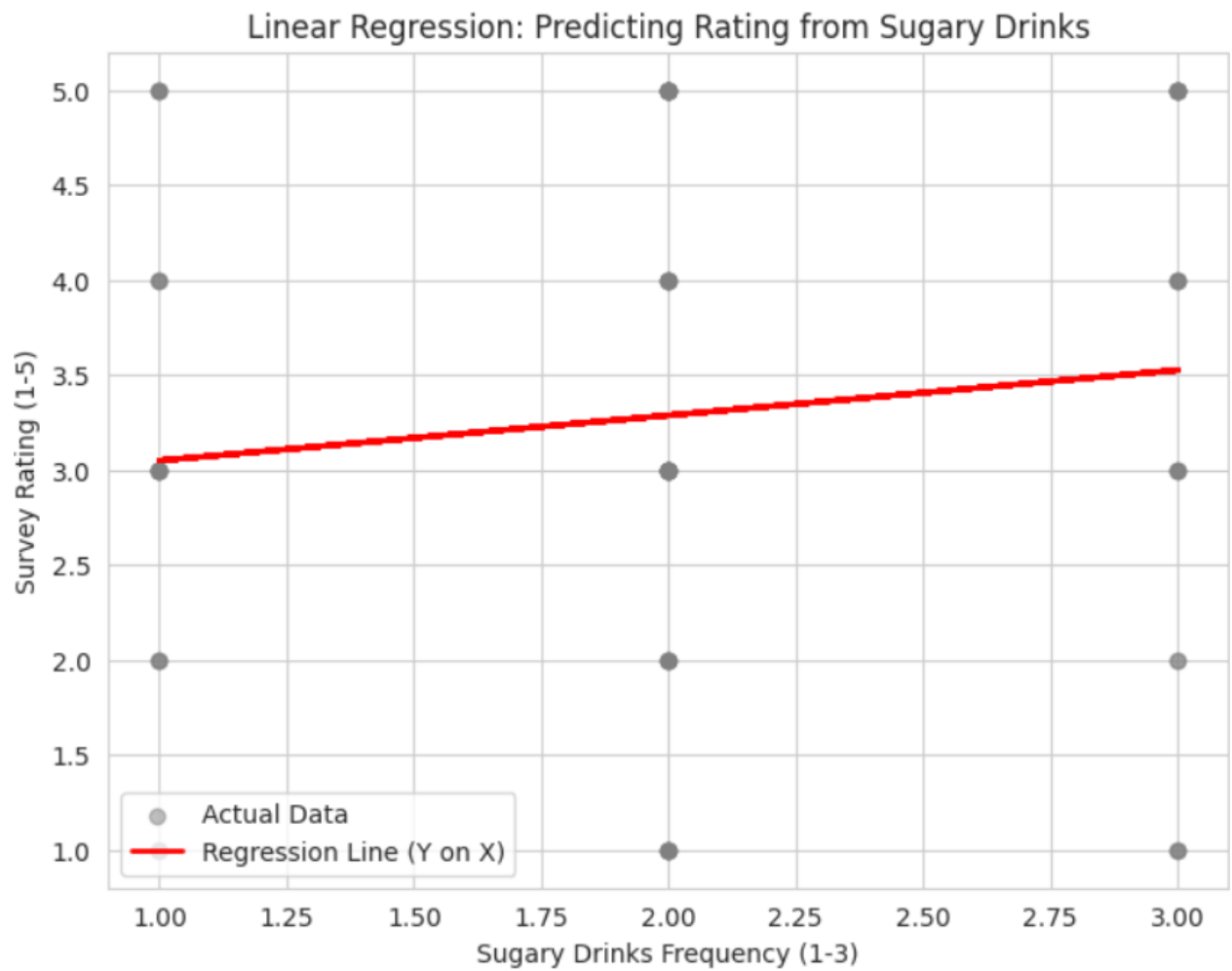
print("LINEAR REGRESSION RESULTS (Y ON X)")
display(regression_table)

# VISUALIZATION
plt.figure(figsize=(8, 6))
plt.scatter(X, Y, color='gray', alpha=0.5, label='Actual Data')
plt.plot(X, model.predict(X), color='red', linewidth=2, label='Regression Line (Y on X)')

plt.title('Linear Regression: Predicting Rating from Sugary Drinks')
plt.xlabel('Sugary Drinks Frequency (1-3)')
plt.ylabel('Survey Rating (1-5)')
plt.legend()
plt.show()
```

❖ Output

box



4. Interpretations

Part A: Dietary Patterns (Q1 - Q3)

Code Analysis (Mode):

- Fruit Consumption: Sometimes (60.9%)
- Vegetable Consumption: Occasionally (51.7%)
- Meals per Day: 3 meals (54.2%)

Interpretation:

The data indicates that most young adults do not consistently consume fruits and vegetables on a daily basis. While the majority maintain a standard pattern of eating 3 meals per day, the irregular intake of fruits and vegetables suggests a gap in nutritional quality. This reflects a moderate level of dietary awareness but inconsistent implementation of healthy eating practices.

Part B: Eating Behavior and Routine (Q4 - Q6)

Code Analysis (Mode):

- Breakfast Skipping: Sometimes (40.6%)
- Fast Food Consumption: Occasionally (54.2%)
- Water Intake: 1–2 liters (48.3%)

Interpretation:

A significant portion of young adults occasionally skip breakfast, indicating irregular eating routines. Fast food consumption on an occasional basis suggests that convenience plays an important role in dietary choices. Although most individuals consume a moderate amount of water, it may still fall short of optimal hydration levels, reflecting an area for improvement in daily health habits.

Part C: Awareness and Decision Factors (Q7 - Q8, Q15)

Code Analysis (Mode):

- Reading Nutrition Labels: Sometimes (45.0%)
- Food Choice Influence: Taste (43.5%)
- Awareness Level: Somewhat aware (47.6%)

Interpretation:

The findings reveal that young adults possess partial awareness about healthy eating

but do not consistently apply it in practice. Taste is the dominant factor influencing food choices, often overriding health considerations. This highlights a behavioral gap where knowledge exists but is not the primary driver of decision-making.

Part D: Unhealthy Consumption Patterns (Q9, Q13, Q14, Q16)

Code Analysis (Mode):

- Sugary Drinks Intake: Occasionally (40.2%)
- Snacking Between Meals: Frequently (48.3%)
- Junk Food Under Stress: Sometimes (45.0%)
- Late Night Eating: Sometimes (40.6%)

Interpretation:

The data suggests that unhealthy eating habits are quite common among young adults, particularly in the form of frequent snacking and emotional eating during stress. Occasional consumption of sugary drinks and late-night eating patterns indicate lifestyle-related challenges. These behaviors are often linked to academic pressure, irregular schedules, and emotional triggers.

Part E: Diet Quality and Preferences (Q10 - Q12, Q17 - Q18)

Code Analysis (Mode):

- Balanced Diet: Sometimes (47.6%)
- Home-Cooked Meals: Daily (51.7%)
- Perception of Healthy Diet: Maybe (45.0%)
- Protein Intake: Sometimes (43.5%)
- Food Preference: Both equally (48.3%)

Interpretation:

Most young adults fall into a middle category where they occasionally follow a balanced diet but lack consistency. While home-cooked meals are common, nutritional balance (especially protein intake) is not always maintained. The preference for both healthy and fast food equally reflects a dual lifestyle, where individuals try to balance health with convenience and taste.

Part F: Physical Activity and Overall Health (Q19 + Combined Insight)

Code Analysis (Mode):

- Exercise Regularly: Sometimes (47.6%)

Interpretation:

The majority of young adults engage in physical activity only occasionally, indicating a lack of consistent fitness routines. When combined with moderate dietary habits, this suggests that overall health management is inconsistent. While there is awareness about maintaining a healthy lifestyle, both diet and exercise lack regular discipline.

5. Statistical Conclusion

The analysis concludes that young adults exhibit moderate awareness of healthy eating but struggle with consistent implementation. Their dietary patterns are influenced heavily by taste, convenience, and lifestyle factors such as stress and time constraints.

Although many individuals attempt to maintain balanced diets and consume home-cooked meals, unhealthy behaviors such as frequent snacking, fast food consumption, and irregular eating schedules persist. Additionally, inconsistent exercise habits further contribute to suboptimal health outcomes.

Overall, there exists a clear **gap between awareness and practice**, highlighting the need for better education, structured routines, and accessible healthy food options to improve the well-being of young adults.